Fatty acids and glucolipotoxicity in the pathogenesis of Type 2 diabetes.

The prevalence of Type 2 diabetes is increasing dramatically as a result of the obesity epidemic, and poses a major health and socio-economic burden. Type 2 diabetes develops in individuals who fail to compensate for insulin resistance by increasing pancreatic insulin secretion. This insulin deficiency results from pancreatic beta-cell dysfunction and death. Western diets rich in saturated fats cause obesity and insulin resistance, and increase levels of circulating NEFAs [non-esterified ('free') fatty acids]. In addition, they contribute to beta-cell failure in genetically predisposed individuals. NEFAs cause beta-cell apoptosis and may thus contribute to progressive beta-cell loss in Type 2 diabetes. The molecular pathways and regulators involved in NEFA-mediated beta-cell dysfunction and apoptosis are beginning to be understood. We have identified ER (endoplasmic reticulum) stress as one of the molecular mechanisms implicated in NEFA-induced beta-cell apoptosis. ER stress was also proposed as a mechanism linking high-fat-diet-induced obesity with insulin resistance. This cellular stress response may thus be a common molecular pathway for the two main causes of Type 2 diabetes, namely insulin resistance and beta-cell loss. A better understanding of the molecular mechanisms contributing to pancreatic beta-cell loss will pave the way for the development of novel and targeted approaches to prevent Type 2 diabetes.